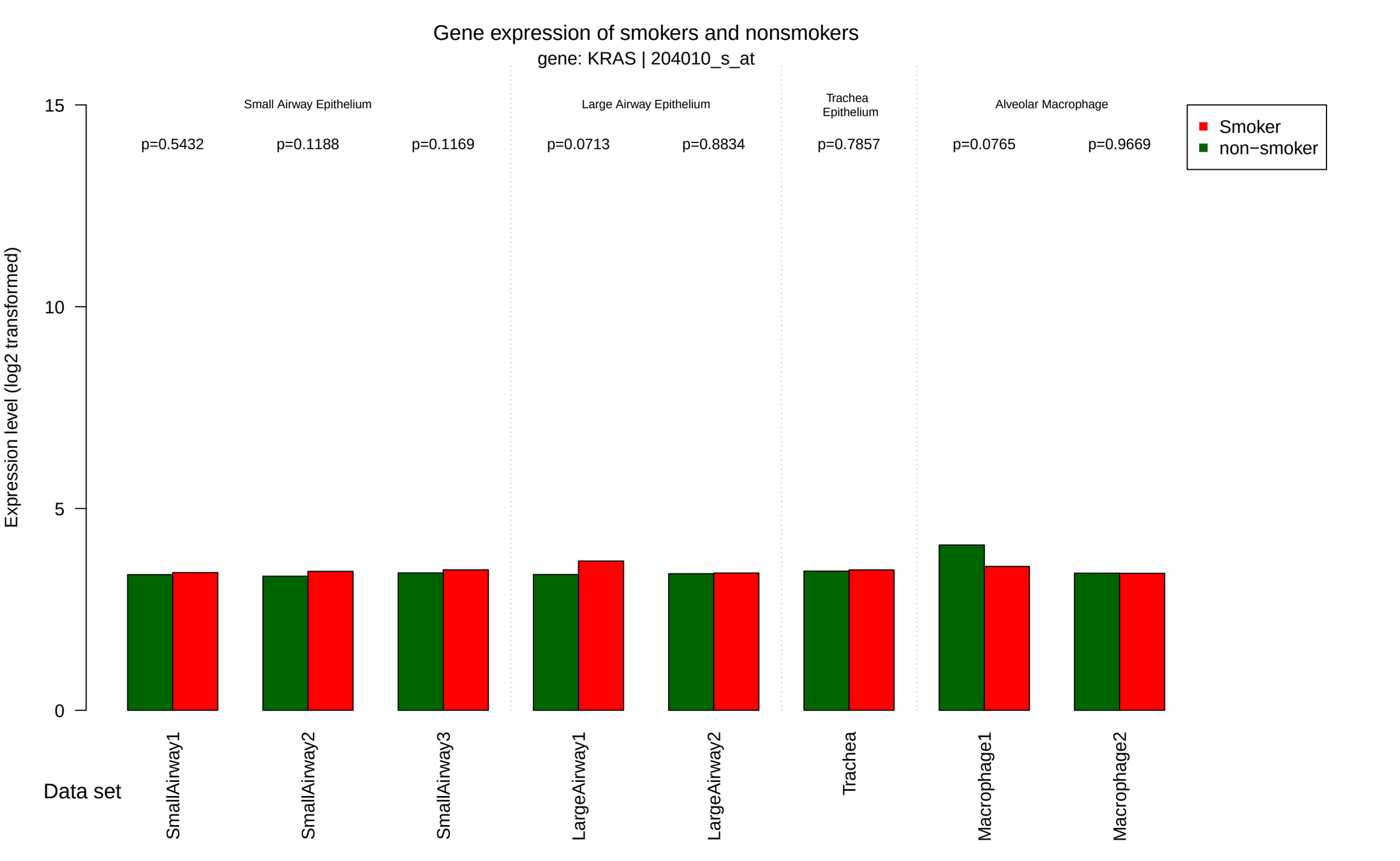
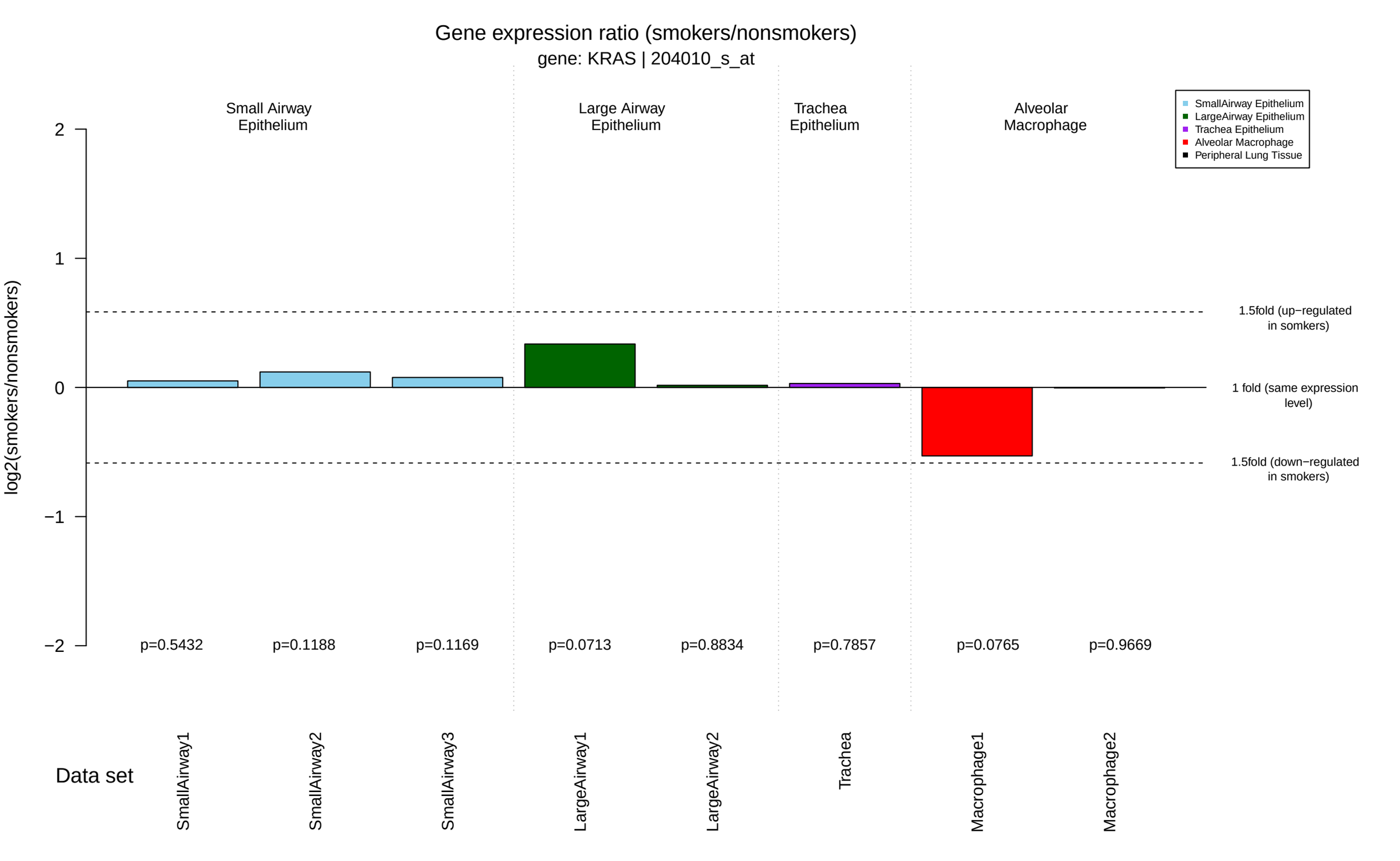
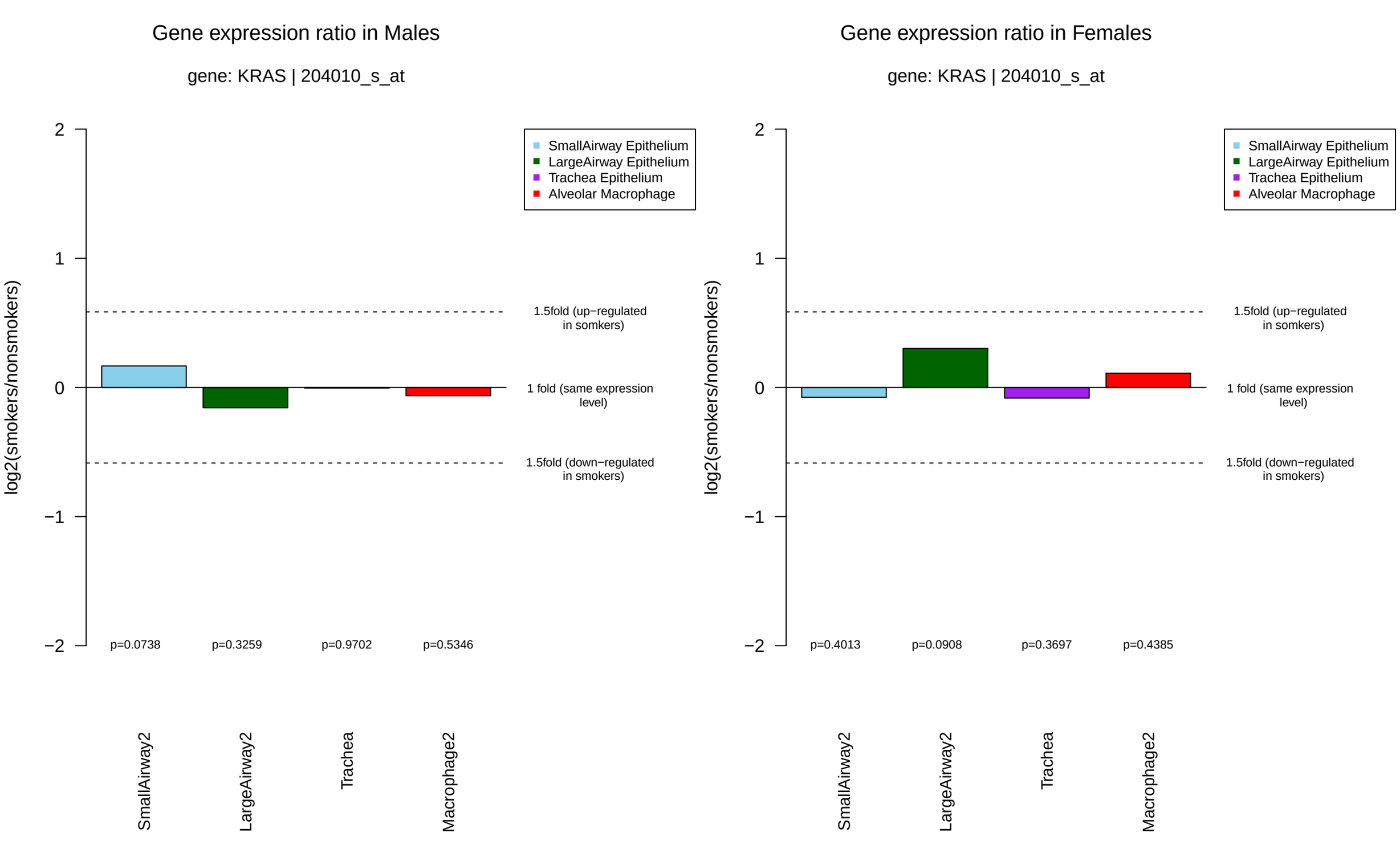
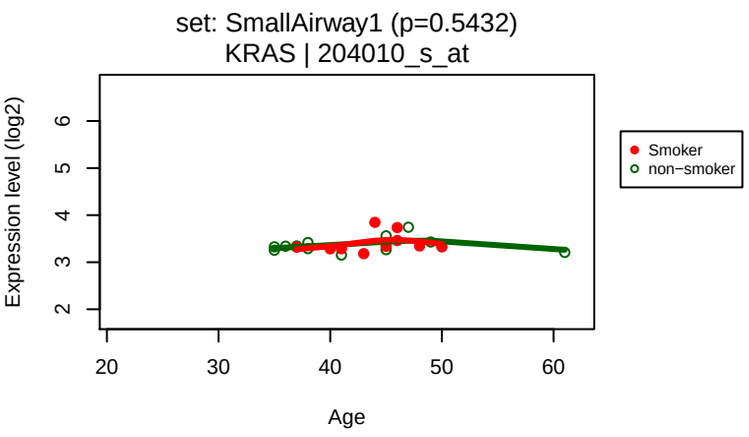
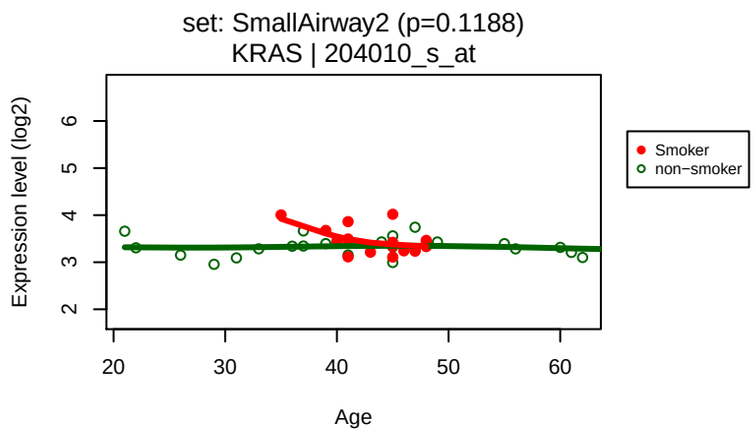
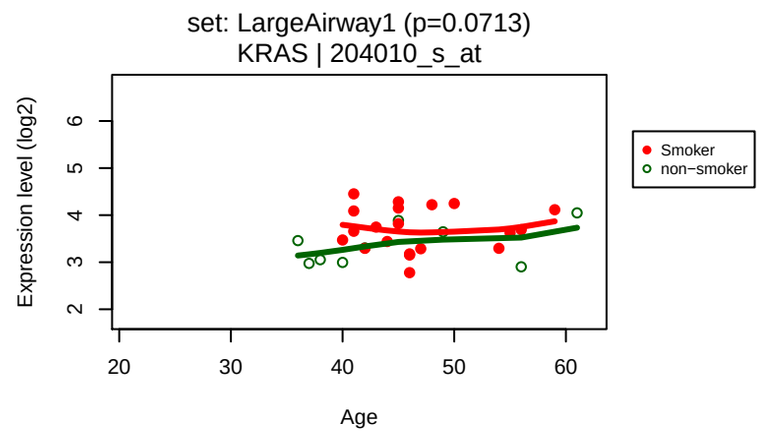

Gene expression of smokers and nonsmokers
gene: KRAS | 204010_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5432
p=0.1188
p=0.1169
p=0.0713
p=0.8834
p=0.7857
p=0.0765
p=0.9669
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

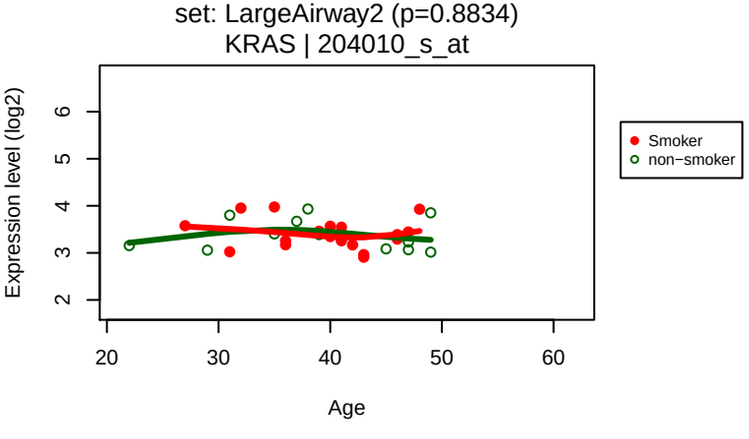

Gene expression ratio (smokers/nonsmokers)
gene: KRAS | 204010_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5432
p=0.1188
p=0.1169
p=0.0713
p=0.8834
p=0.7857
p=0.0765
p=0.9669
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

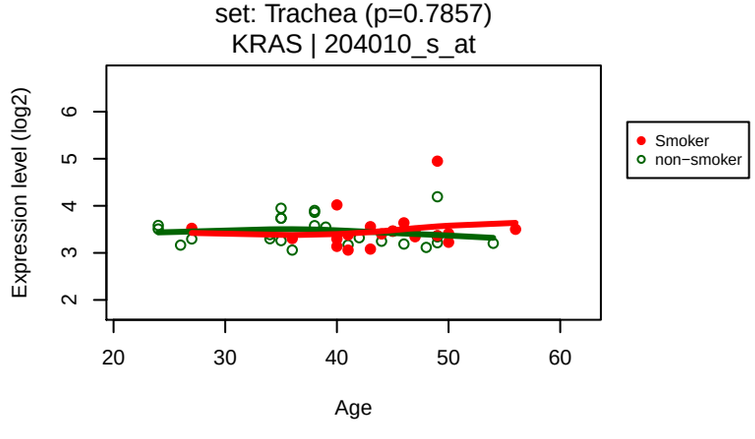

Gene expression ratio in Males
gene: KRAS | 204010_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0738
p=0.3259
p=0.9702
p=0.5346
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KRAS | 204010_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4013
p=0.0908
p=0.3697
p=0.4385
SmallAirway2
LargeAirway2
Trachea
Macrophage2

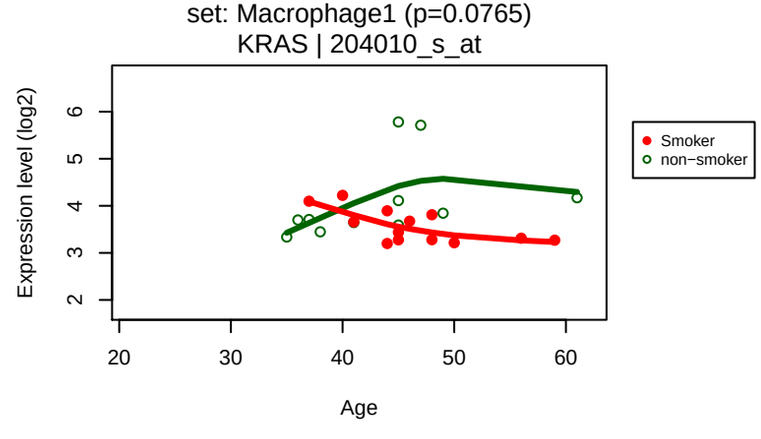

set: SmallAirway1 (p=0.5432)
KRAS | 204010_s_at
Expression level (log2)
Age
Smoker
non−smoker

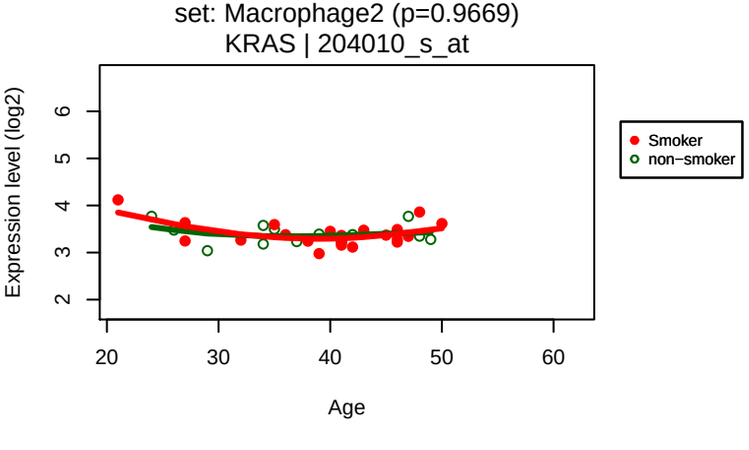

set: SmallAirway2 (p=0.1188)
KRAS | 204010_s_at
Expression level (log2)
Age
Smoker
non−smoker

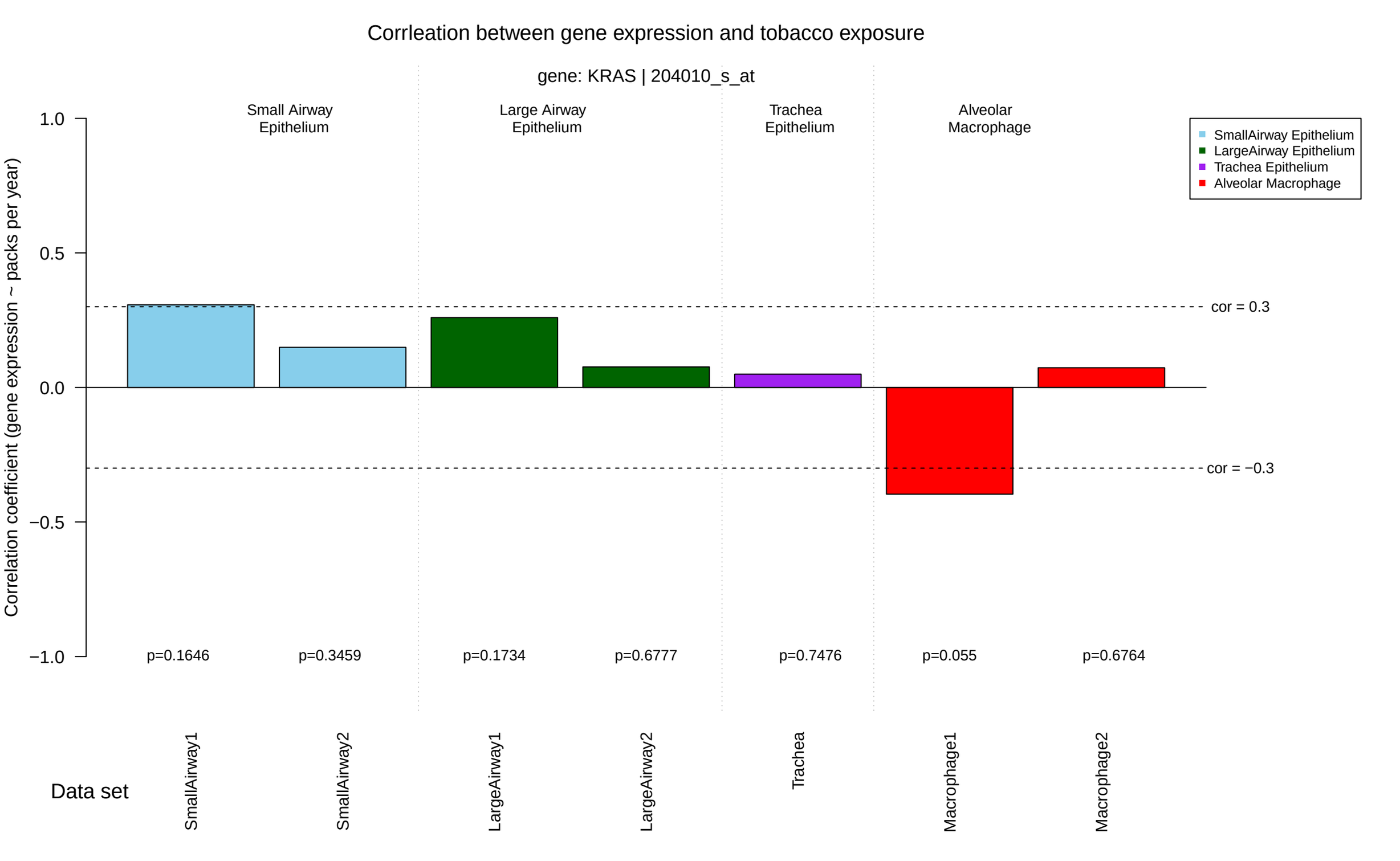

set: LargeAirway1 (p=0.0713)
KRAS | 204010_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8834)
KRAS | 204010_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7857)
KRAS | 204010_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0765)
KRAS | 204010_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9669)
KRAS | 204010_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KRAS | 204010_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1646
p=0.3459
p=0.1734
p=0.6777
p=0.7476
p=0.055
p=0.6764
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2